COMP3009 Machine Learning Assignment 1: Classification and Regression

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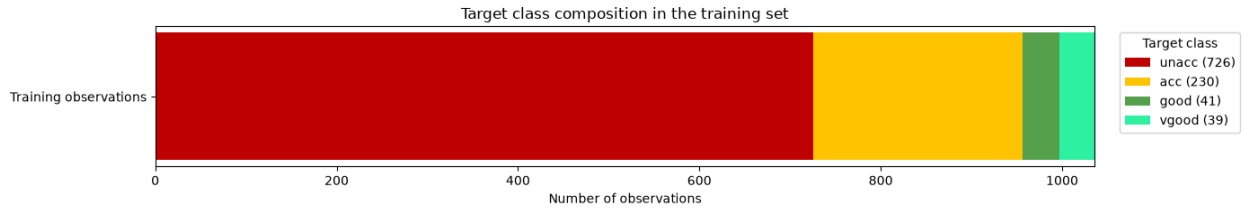
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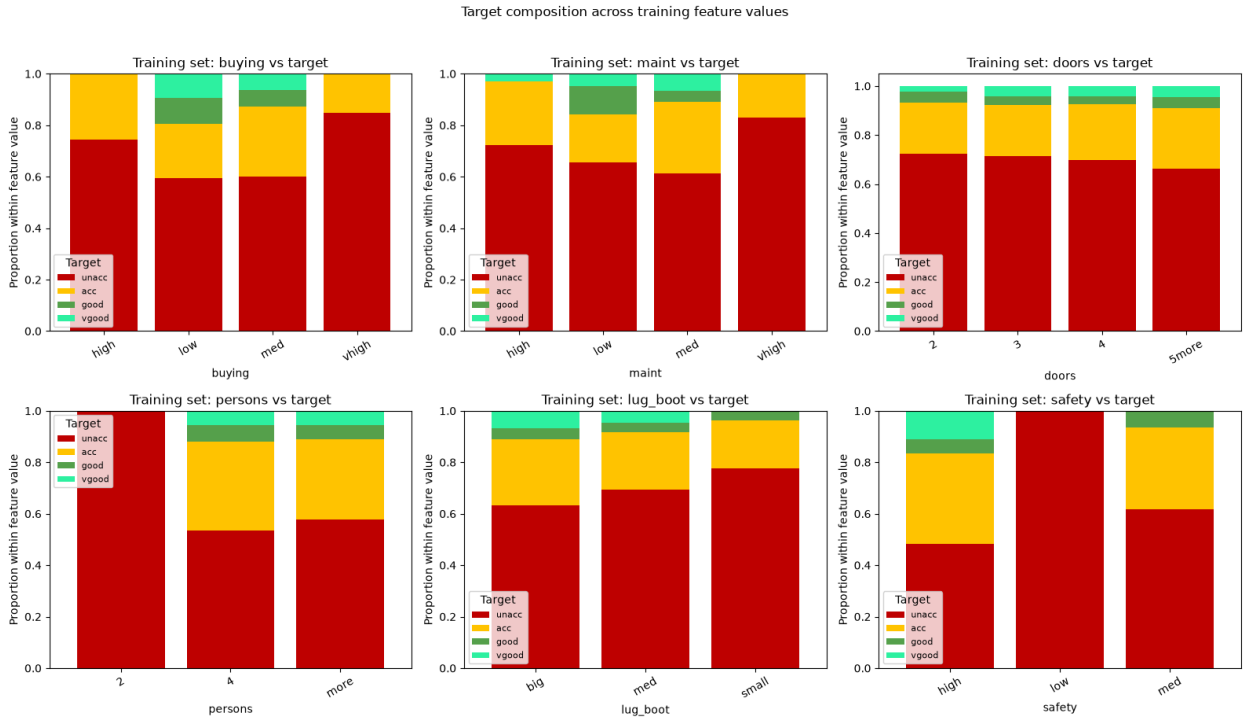
Introduction

This report applies four machine learning methods to two UCI datasets: classifying car acceptability and predicting hourly carbon monoxide concentration. Linear and logistic regression fit weighted feature combinations. Support vector machines fit maximum-margin models, using kernels for non-linear patterns. Decision trees split data recursively on features. Multi-layer perceptrons learn non-linear mappings through hidden layers.

Classification Results

The classification task used the UCI Car Evaluation dataset to predict different acceptability classes to determine the state of a car. There are: unacc (Unacceptable), acc (Acceptable), good (Good), and vgood (Very Good). The dataset contains 1728 observations and six features: buying price, maintenance price, number of doors, passenger capacity, luggage boot size, and car safety. In an initial check of the database, there are no missing values, duplicate rows, unexpected feature values, or missing or conflicting labels. However, the dataset's target class was imbalanced. With 70.0% of cars considered unacc, 22.2% acc, 4.0% good, and 3.8% vgood. Due to this, we determine that macro F1 and classification accuracy will be good metrics to have as it gives each class equal weight.





For training the models, we used a 60/40 split, with 1036 training and 692 test observations respectively. Each model used a pipeline to one-hot encode categorical variables before classifications. We then used four models: Logistic Regression (C = 10.0), Support Vector Machine (C = 10.0, gamma = scale, kernel = rbf), Decision Tree (max_depth = 15, min_samples_leaf = 1), and Multi-Layer Perceptron (alpha = 0.0001, hidden_layer_sizes = (100, 50), learning_rate_init = 0.01). These hyperparameters were selected with the help of grid search that used macro F1 as the selection score. A nested five-fold cross-validation on the training data to evaluate the models was also done.

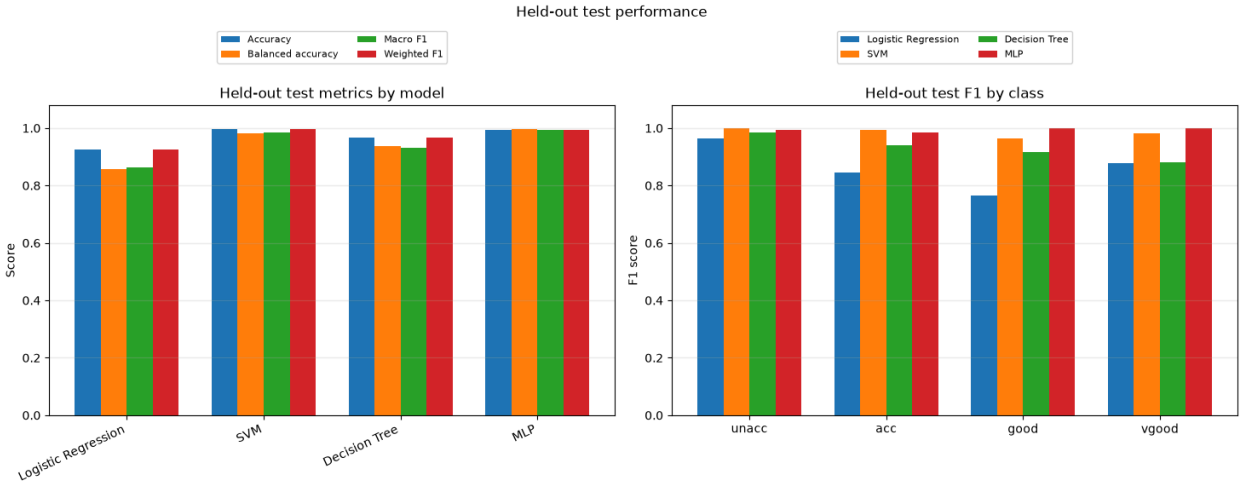
In the training split, SVM achieved the highest nested CV accuracy of 98.75% and a macro F1 of 0.964. The runner-up is MLP, with a CV accuracy of 98.74% and a macro F1 of 0.962. Decision Tree achieved a 95.37% accuracy and 0.875 macro F1. The model that performed worse is Logistic Regression, with an accuracy of 91.99% and macro F1 of 0.852. Lastly, the small accuracy standard deviation values of 0.004932-0.016964 indicate relatively consistent results across the outer folds. However, the macro F1 standard deviation is much more varied, with the Decision Tree with the highest standard deviation of 0.042.

Model	Nested CV macro F1	Macro F1 std	Nested CV accuracy	Accuracy std
SVM	0.963899	0.028917	0.987467	0.009901
MLP	0.962497	0.019981	0.987449	0.004932
Decision Tree	0.875158	0.041726	0.953670	0.007218
Logistic Regression	0.852042	0.022653	0.919867	0.016964

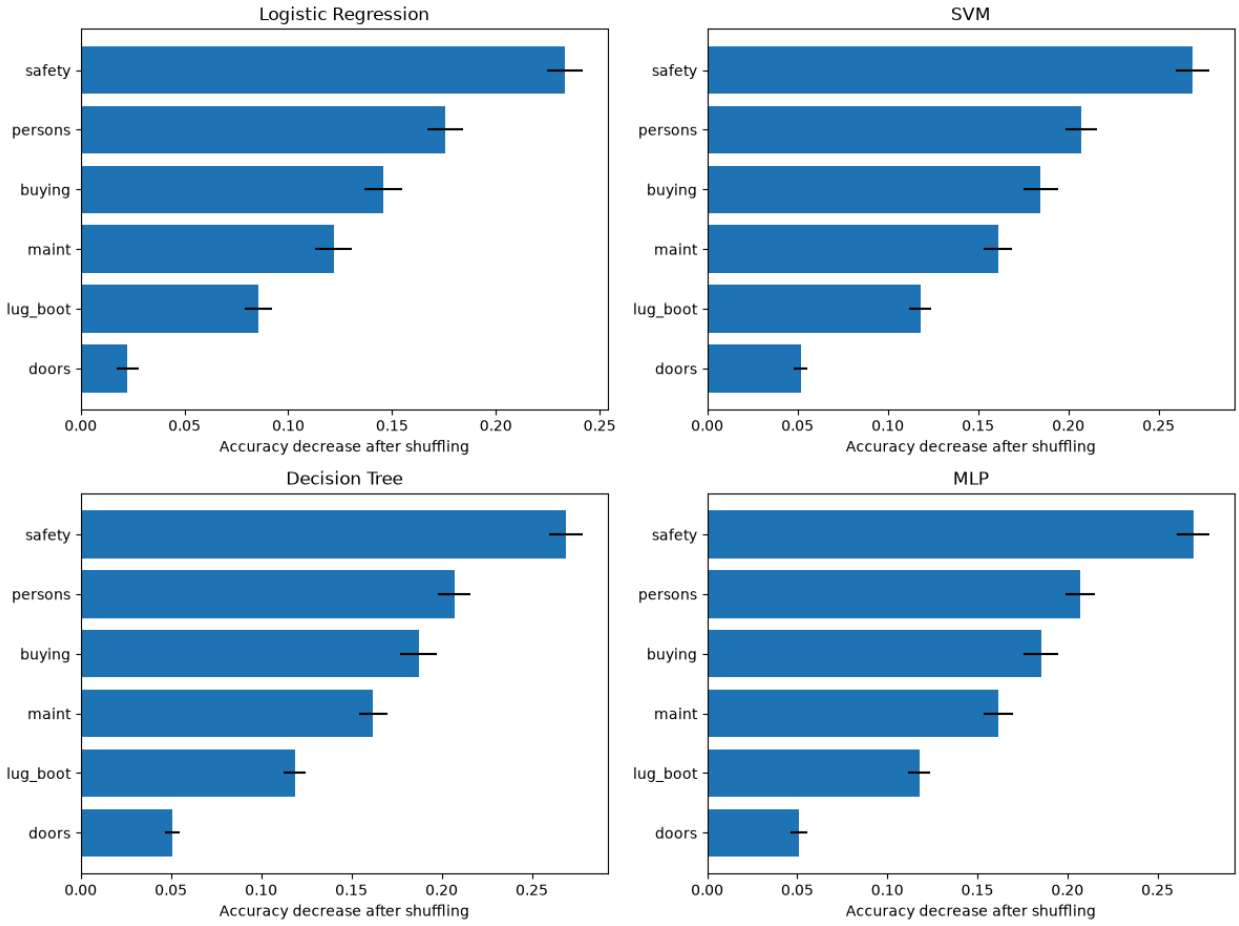


Followed up by the results of the held-out test, it follows the same trend of results. SVM achieved the best accuracy of 99.57%, followed by MLP with 99.28%, Decision Tree with 96.82%, and lastly Logistic Regression with 92.63%. However, for the highest test macro F1, instead of SVM, MLP achieved the highest score of 0.9946, followed by SVM with 0.9842, Decision Tree with 0.9300, and Logistic Regression with 0.8630.

Model	Test accuracy	Test balanced accuracy	Test macro F1	Test weighted F1
MLP	0.992775	0.996311	0.994685	0.992798
SVM	0.995665	0.980940	0.984298	0.995659
Decision Tree	0.968208	0.936081	0.930000	0.968375
Logistic Regression	0.926301	0.856087	0.863029	0.925667

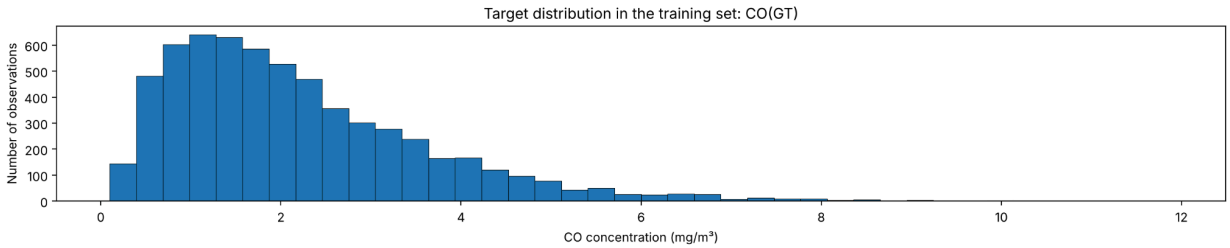


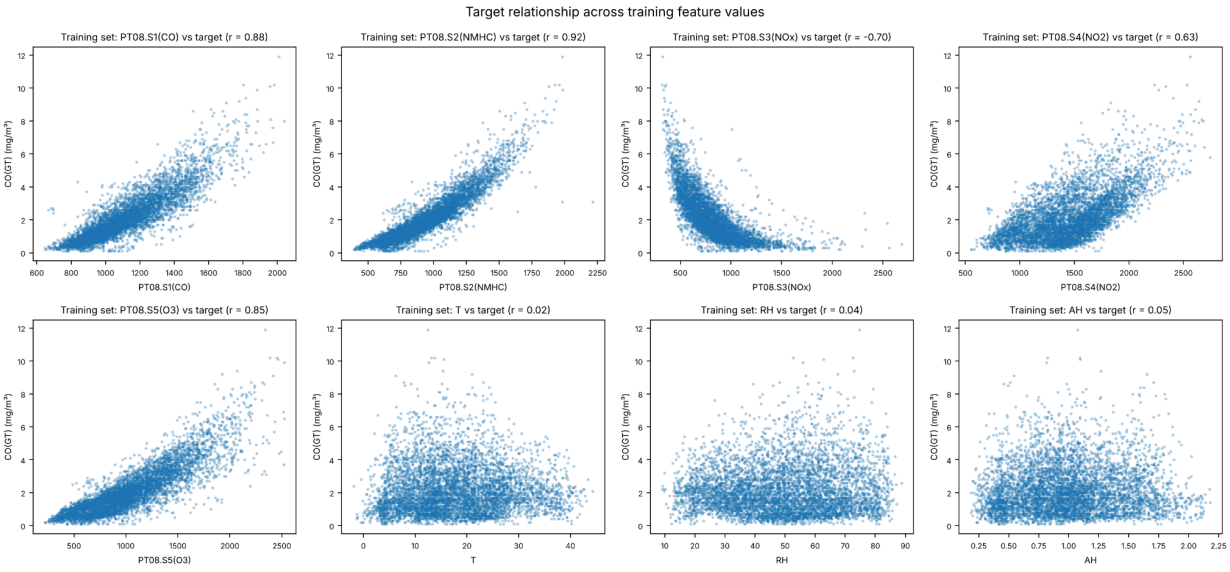
Lastly, we conducted a chi-square feature-selection comparison and identified that using all features produced better results rather than selecting only four or five for the training data. We wanted to ensure all features contributed to improving results rather than making noise as shown in the graph.



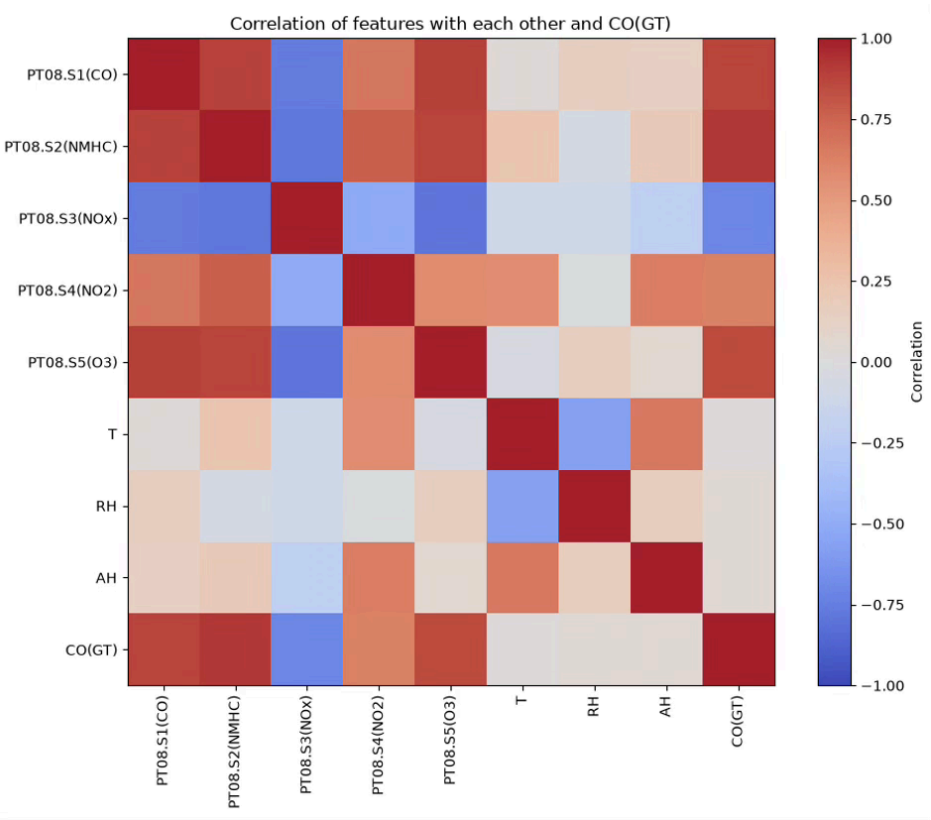
Regression Results

The regression task used the UCI Air Quality dataset to predict hourly carbon monoxide concentration. The raw data contains 9,357 hourly readings. Missing values were coded as -200, so these were replaced with NaN. The 1,683 rows with no CO value were also removed, leaving 7,674 observations. The eight predictors that were used in testing were the five metal-oxide sensor responses, temperature, relative humidity, and absolute humidity. For the dataset, NMHC was excluded because 90.2% of its values were missing. C6H6, NOx and NO2 were also excluded because they have the same reference measurements as the target. Date and time were also excluded. In the dataset, the target was right-skewed, with a training median of 1.8 mg/m^3 and a maximum of 11.9 mg/m^3.





The correlation heatmap (Figure X) shows that PT08.S2(NMHC), PT08.S1(CO) and PT08.S5(O3) had strong positive correlations with CO (0.92, 0.88 and 0.85), while PT08.S3(NOx) had a strong negative correlation (-0.70). Temperature and humidity had almost no linear correlation with CO. Several relationships are curved, suggesting non-linear models may fit better.



Using an 80/20 train-test split, our models produced 6,139 training and 1,535 test observations. Each model used a pipeline to fill missing values with the training median and then standardised the features. There were four methods that were compared: Linear Regressions, Support Vector Regression (SVR, RBF kernel, C = 1.0, ε = 0.1), Decision Tree (max_depth = 10) and Multi-Layer Perceptron (MLP, one hidden layer of 100 ReLU units, Adam, max_iter = 1000). Fixed settings were used instead of grid search and models were evaluated with shuffled five-fold cross-validation on the training data, using MSE as the main metric and R^2 as a second metric. The held-out test set was evaluated once, after fitting each model on all training data.

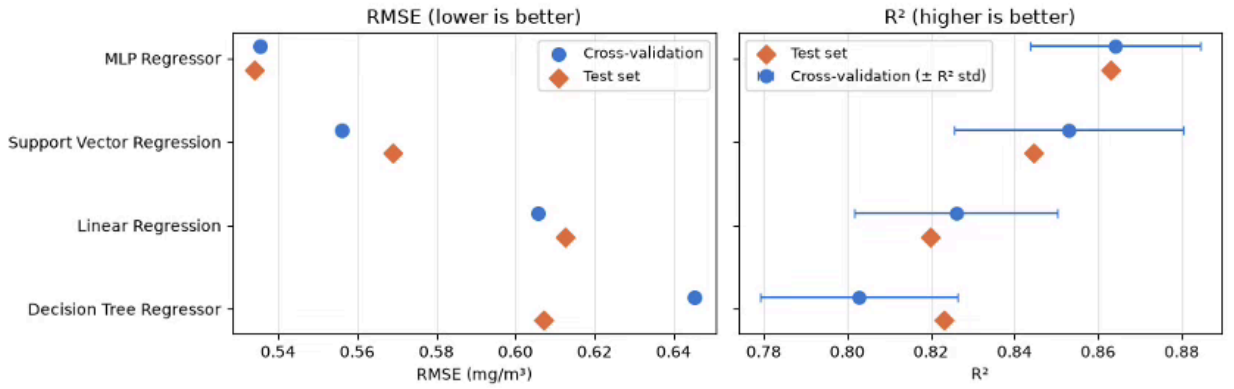
What was found was that in cross-validation, MLP had the lowest mean MSE of 0.2888 and the highest R^2 of 0.8640. SVR followed with an MSE of 0.3128. Linear Regression scored 0.3695 and Decision Tree 0.4184. The MSE (0.0513–0.0673) and R^2 (0.0203–0.0273) standard deviations show similar variation across folds for all four models.

Model	CV MSE	MSE std	CV R^2	R^2 std
MLP	0.2888	0.0513	0.8640	0.0203
SVR	0.3128	0.0673	0.8529	0.0273
Linear Regression	0.3695	0.0626	0.8260	0.0243
Decision Tree	0.4184	0.0623	0.8028	0.0236

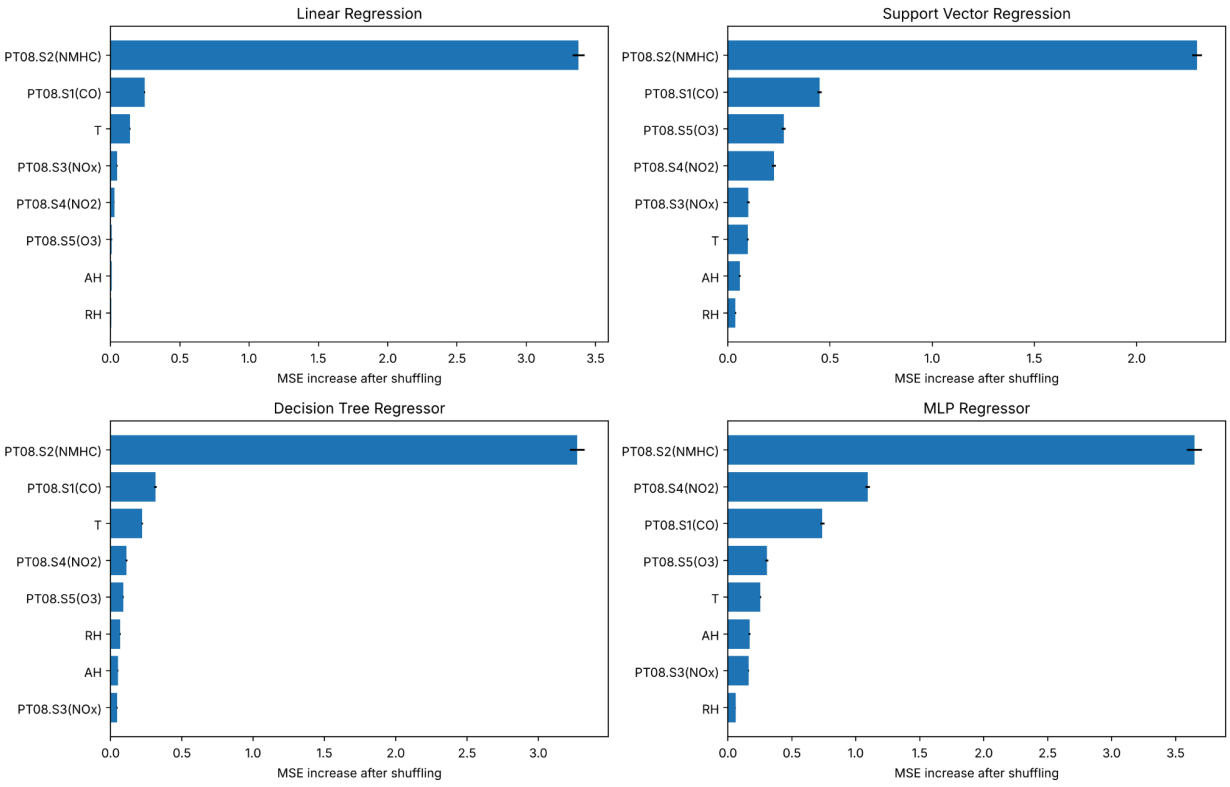
Model	CV MSE	CV RMSE	CV R^2	R^2 std
MLP	0.2888	0.5353	0.8640	0.0203
SVR	0.3128	0.5560	0.8529	0.0273
Linear Regression	0.3695	0.6057	0.8260	0.0243
Decision Tree	0.4184	0.6451	0.8028	0.0236

The results supported the cross-validation findings. MLP reached a test MSE of 0.2852 (R^2 = 0.8631), and SVR reached 0.3237. Decision Tree scored 0.3684 and Linear Regression 0.3753. Decision Tree did slightly better than Linear Regression on the test set, which reverses their cv order. This shows the gap between those two models is small. SVR had the lowest test MAE (0.2961), but its larger errors gave it a higher MSE than MLP.

Model	Test MSE	Test RMSE	Test MAE	Test R^2
MLP	0.2852	0.5340	0.3062	0.8631
SVR	0.3237	0.5689	0.2961	0.8446
Decision Tree	0.3684	0.6070	0.3636	0.8231
Linear Regression	0.3753	0.6126	0.3645	0.8198



Test-set permutation importance ranked PT08.S2(NMHC) highest for all four models. PT08.S1(CO) ranked second for most, while MLP also relied heavily on PT08.S4(NO2). Humidity had little effect.



Overall, MLP gave the strongest regression results, followed by SVR. Because the readings are hourly, shuffled folds place neighbouring hours in both training and validation sets, so the scores may be slightly optimistic.

Conclusion

For the classification task, SVM and MLP performed the best. After that, there is a noticeable drop in performance for Decision Tree and Logistic Regression, with Logistic Regression receiving the lowest scores for its metrics. Lastly, there is possible overfitting for SVM, MLP, and Decision Tree due to their high training macro F1 scores and lower CV scores. For the regression task, MLP performed best for regression (test MSE 0.2852), followed by SVR (0.3237). The non-linear SVM, SVR and MLP outperformed linear and logistic regression on both tasks, suggesting non-linear relationships in the data.